

SWAPCELL · DESIGN DECISION RECORD

SwapCell recommendations accepted

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DOCUMENT CONTROL

Document ID	SWC-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/swapcell		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all remaining recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted

■ Context

After SWC-DDR-001, seven items were still listed as "Proposed, awaiting Amish" (SWC-DDR-001 items 14 to 20, docs/REVIEW.md session 2026-09-25). On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every open item that carried a recommendation is therefore decided, with the recommended option as the decision. Items with no recommendation stay open. TRL 4 remains on hold by Amish's instruction, so decisions that need a build, a test or a purchase are recorded but not carried out.

■ Options considered

The options for each item are in SWC-DDR-001 Table 2, SWC-PRC-001 v0.3 and SWC-CAL-001 v0.1. They are not repeated here.

■ Decision

Table 1. Newly decided items. Each is "Decided by Amish, 2026-09-25: go with recommendation".

# (DDR-001)	Item	Decision	What changed in the repo
17	Connector family	Custom printed keyed shroud around commercial high-current socket contacts and potted spring signal contacts	SWC-PRC-001 v0.4 components table, design choices and open questions; SWC-REQ-001 v0.4 R10 verification note; bom/bom.csv item 5 note; SWC-CAL-001 v0.2 R10 value; SWC-DWG-002 note added (Rev P1 to P2). No geometry change: the model already showed a keyed shroud. Contact part selection and R8 and R10 verification are TRL 4 work, on hold
19	Values inside interface v0.3 items W, C and V	Confirmed as proposed: 10 kΩ coded INTERLOCK, 100 μA sleep limit, 25 g shock, 330 N receiver preload, charge-FET fallback on a charge-side fault	SWC-PRC-001 v0.4 interface note; SWC-REQ-001 v0.4 assumptions; SWC-CAL-001 v0.2 inputs table. Numbers unchanged
20	R3 thermal rating	Keep 20 A continuous at 25 °C with temperature derating in PACK_LIMITS, rather than a 15 A label	R3 restated in SWC-REQ-001 v0.4 (20 A in an open-air mount from 25 °C; derating elsewhere). New derating rule in SWC-PRC-001 v0.4: full current to 50 °C cell temperature, linear to 5 A at 60 °C; 35 A peak only below 50 °C. sizing.py now also computes the current that holds 60 °C enclosed (about 18.5 A) beside the 45 °C ambient case (about 14.0 A). R3 status: at risk to met on paper. SWC-DWG-002 note added
15	Build budget for a second pack	Keep budget_usd: 700 for TRL 3 and set the build budget (about \$973 for two packs and one dock) before any build	No change to project.yaml. Decided but on hold: purchasing is TRL 4 work. SWC-REQ-001 v0.4 R16 assumption and bom/bom-notes.md updated

Items still open

Table 2. Items that stay Proposed, awaiting Amish (no recommendation to accept).

# (DDR-001)	Item	Why still open
14	First co-design partner (repair workshop, delivery fleet or rural e-bike cooperative)	Portfolio rule: partners are chosen per area later; no choice to accept
16	LFP variant (16S)	No preference stated
18	EnergyBus gateway mapping	Needs the CiA 454 specification, which was not read; no recommendation to accept

■ Consequences

- Requirement status (SWC-CAL-001 v0.2): not met, none; at risk, 2 (R2, R9; was 3); not verifiable at TRL 3, 5 (R7, R8, R10, R11, R15); met, 9 (R1, R3, R4, R5, R6, R12, R13, R14, R16; four on paper only).
- `budget_usd` stays at \$700; the priced BOM is \$559.
- SwapCell interface stays at v0.3. The derating rule is pack behavior inside PACK_LIMITS and needs no change in any receiver; the confirmed values match what dependent repos already cite.
- No cross-repo edits are needed. Dependent repos may note that the v0.3 values are now confirmed and that vehicle receivers should still leave the back and lid faces open to air to keep the full 20 A.
- `trl: 3` and `trl_target: 3` are unchanged. No TRL 4 work was started.